

Decision Control Structures in C++ (if-else, switch, ternary)

> **Definition to Remember**

> Dec
ternary

1. Concept Overview & Fundamentals

Decision Control Structures in C++ (if-else, switch, ternary) provides modern high-performance capabilities combining low-level hardware control with high-level abstractions, object-oriented paradigms, and Standard Template Library (STL) utilities.

In modern C++ (C++11 through C++20):

- **Z

2. Technical Architecture & Memory Mechanics

C++ Runtime Execution Architecture

```text

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### Modern C++ Feature Comparison

| Modern C++ Paradigm | Legacy Equivalent | Engineering Advantage | Performance Impact |

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3. Core Syntax & Implementation Patterns

Modern C++ Template Structure

```
```cpp
// Fast I/O and standard namespace configuration

std::cout << "Executing C++ Module: Decision Control Structure (if-else, switch, goto, ternary)" << "\n";

return 0;
```
```

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4. Real-World Applications & System Engineering

- **Quantitative Finance & HFT:** High-frequency trading platforms use modern C++ for microsecond-level execution.

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5. Comprehensive Solved Problems & Code Walkthroughs

Problem 1: AbsoluteTO

**Problem Description & Objective:**

**C++ Code Implementation:**

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream> // Header for console stream I/O
```

```
using namespace std; // Standard namespace
```

```
// --- Main Execution Entry Point ---
```

```
int main
```

```
// Prompt user for input integer
```

```
cout <<
```

```
// Evaluate absolute value using ternary conditional operator
```

```
// Con
```

```
// Output calculated absolute value
```

```
cout <<
```

```
// Return zero
```

```
return
```

```
---
```

Problem 2: CategorizeAgeGroup

****Problem Description & Objective:****

Imple

****C++ Code Implementation:****

```cpp

/\*

- ===

```
#include <iostream> // Console I/O stream header
```

```
using namespace std; // Standard namespace
```

```
// --- Main Execution Entry Point ---
```

int ma

```
// Prompt for user age
```

cout <

```
// Branch 1: Invalid negative age check
```

if (age

```
// Return 0 indicating clean finish
```

```
return
```

```

```

### Problem 3: GradeCalculator

**\*\*Problem Description & Objective:\*\***

Implemen

**\*\*C++ Code Implementation:\*\***

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Output formatted data to console stream
```

```
// Evaluate conditional decision logic
```

```
// Return zero code indicating successful program finish
```

```
---
```

Problem 4: GreatestNumber

****Problem Description & Objective:****

****C++ Code Implementation:****

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
using namespace std;
```

```
// --- Main Execution Entry Point ---
```

```
int main()
```

```
// Output formatted data to console stream
```

```
cout << endl;
```

```
// Evaluate conditional decision logic
```

```
if (n1 > n2)
```

```
// Return zero code indicating successful program finish
```

```
return 0;
```

Problem 5: LeapYear

****Problem Description & Objective:****

****C++ Code Implementation:****

/*

#include <iostream>

// Use standard library namespace globally

// --- Main Execution Entry Point ---

// Output formatted data to console stream

// Evaluate conditional decision logic

```
// Return zero code indicating successful program finish
```

```
return
```

```
---
```

Problem 6: MinimumNumberTO

****Problem Description & Objective:****

Implemen

****C++ Code Implementation:****

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
using n
```

```
// --- Main Execution Entry Point ---
```

```
int mai
```

```
// Output formatted data to console stream
```

```
cout <
```

```
int minVal = (a < b) ? a : b;
```

```
// Output formatted data to console stream
```

```
cout <
```

```
// Return zero code indicating successful program finish
```

```
return
```

```

```

### Problem 7: NumberChecker

**\*\*Problem Description & Objective:\*\***

Implem

**\*\*C++ Code Implementation:\*\***

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
using n
```

```
// --- Main Execution Entry Point ---
```

```
// Output formatted data to console stream
```

```
// Evaluate conditional decision logic
```

```
// Return zero code indicating successful program finish
```

```
---
```

Problem 8: OddEven

****Problem Description & Objective:****

****C++ Code Implementation:****

```
/*
```

```
#include <iostream>

// Use standard library namespace globally

// --- Main Execution Entry Point ---

// Output formatted data to console stream

// Evaluate conditional decision logic

// Return zero code indicating successful program finish
```

```
---
```

Problem 9: OddEvenTO

****Problem Description & Objective:****

****C++ Code Implementation:****

```
/*
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Output formatted data to console stream
```

```
string result = (number % 2 == 0) ? "Even" : "Odd";
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```
---
```

Problem 10: PrintMonthSwitch

****Problem Description & Objective:****

****C++ Code Implementation:****

```
/*
```

```
#include <iostream> // Header for standard console stream I/O
```

```
using namespace std; // Standard namespace
```

```
// --- Main Execution Entry Point ---
```

int main

```
// Read month number input from user
```

cout <<

```
// Switch statement jumps directly to matching integer case
```

switch

```
// Finish program execution
```

```
return
```

```
---
```

Problem 11: ScoreCategoryTO

****Problem Description & Objective:****

Implem

****C++ Code Implementation:****

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
#includ
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Output formatted data to console stream
```

```
string category = (score > 80) ? "High" : ((score >= 50) ? "Moderate" : "Low");
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```

```

## Problem 12: SimpleCalculatorSwitch

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

```
/*
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
using namespace std;
```

```
// --- Main Execution Entry Point ---
```

```
int main()
```

```
// Output formatted data to console stream
```

```
cout << endl;
```

```
// Output formatted data to console stream
```

```
cout << endl;
```

```
// Switch jump-table selection statement
```

```
switch (choice)
```

```
// Return zero code indicating successful program finish
```

return

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## 6. Exam Scoring Points & Memory Revision

> **Must-Write Points**

> 1. sv

> **Quick Recall**

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